

Core Healthcare Dataset

Data Dictionary

10 tables covering patients, organizations, providers, payers, encounters, and core clinical events (conditions, procedures, medications, observations, careplans).

How This Dictionary Is Organized

Each table below corresponds to one entry in the Core ER Diagram. Every entry lists the source file, row/column counts, primary key, foreign keys to other tables, and a full column-by-column listing with data type and description.

Table Groups

Reference tables — PATIENTS, ORGANIZATIONS, PROVIDERS, PAYERS

Encounter hub — ENCOUNTERS

Clinical event tables — CONDITIONS, PROCEDURES, MEDICATIONS, OBSERVATIONS, CAREPLANS

Notes on Data Types

Data types reflect how pandas inferred each column from the raw CSV. Columns typed "float" that hold reference codes (e.g., REASONCODE) are numeric only because of missing values (NaN) in many rows — the codes themselves are whole numbers (SNOMED-CT, RxNorm, LOINC). UUID-typed columns are stored as plain strings in the CSVs.

1. PATIENTS

One row per synthetic patient. Central demographic record referenced by ENCOUNTERS via PATIENT.

Source file: patients.csv

Rows × Columns: 1,163 × 25

Primary Key: Id

Foreign Keys: None

Column	Data Type	Description
Id	<i>string (UUID)</i>	Primary key. Unique patient identifier.
BIRTHDATE	<i>date</i>	Patient date of birth.
DEATHDATE	<i>date</i>	Patient date of death; blank if still alive.
SSN	<i>string</i>	Simulated Social Security Number.
DRIVERS	<i>string</i>	Simulated driver's license number.
PASSPORT	<i>string</i>	Simulated passport number.
PREFIX	<i>string</i>	Name prefix (e.g., Mr., Mrs.).
FIRST	<i>string</i>	First name.
LAST	<i>string</i>	Last name.
SUFFIX	<i>string</i>	Name suffix (e.g., Jr., III).
MAIDEN	<i>string</i>	Maiden name, if applicable.
MARITAL	<i>string</i>	Marital status code (M = Married, S = Single).
RACE	<i>string</i>	Race category.
ETHNICITY	<i>string</i>	Ethnicity category.
GENDER	<i>string</i>	Gender (M/F).
BIRTHPLACE	<i>string</i>	City/state/country of birth.
ADDRESS	<i>string</i>	Home street address.
CITY	<i>string</i>	Home city.
STATE	<i>string</i>	Home state.
COUNTY	<i>string</i>	Home county.
ZIP	<i>float</i>	Home ZIP code.
LAT	<i>float</i>	Latitude of home address.
LON	<i>float</i>	Longitude of home address.
HEALTHCARE_EXPENSES	<i>float</i>	Lifetime total healthcare expenses billed.
HEALTHCARE_COVERAGE	<i>float</i>	Lifetime total amount covered by insurance.

2. ORGANIZATIONS

Healthcare facilities (hospitals, clinics, labs) that employ providers.

Source file: organizations.csv

Rows × Columns: 1,127 × 11

Primary Key: Id

Foreign Keys: None

Column	Data Type	Description
Id	<i>string (UUID)</i>	Primary key. Unique organization identifier.
NAME	<i>string</i>	Organization name.
ADDRESS	<i>string</i>	Street address.
CITY	<i>string</i>	City.
STATE	<i>string</i>	State.
ZIP	<i>string</i>	ZIP code.
LAT	<i>float</i>	Latitude.
LON	<i>float</i>	Longitude.
PHONE	<i>string</i>	Contact phone number.
REVENUE	<i>float</i>	Total revenue generated by this organization.
UTILIZATION	<i>int</i>	Count of encounters handled by this organization.

3. PROVIDERS

Individual clinicians who deliver care during encounters, each affiliated with one organization.

Source file: providers.csv

Rows × Columns: 5,056 × 12

Primary Key: Id

Foreign Keys: ORGANIZATION → ORGANIZATIONS.Id

Column	Data Type	Description
Id	<i>string (UUID)</i>	Primary key. Unique provider identifier.
ORGANIZATION	<i>string (UUID)</i>	Foreign key to ORGANIZATIONS.Id — employing facility.
NAME	<i>string</i>	Provider full name.
GENDER	<i>string</i>	Provider gender (M/F).
SPECIALITY	<i>string</i>	Clinical specialty (e.g., GENERAL PRACTICE).
ADDRESS	<i>string</i>	Practice street address.
CITY	<i>string</i>	Practice city.
STATE	<i>string</i>	Practice state.
ZIP	<i>string</i>	Practice ZIP code.
LAT	<i>float</i>	Latitude.
LON	<i>float</i>	Longitude.
UTILIZATION	<i>int</i>	Count of encounters performed by this provider.

4. PAYERS

Insurance companies (payers), including Medicare/Medicaid and 'NO_INSURANCE', with aggregate financial and enrollment statistics.

Source file: payers.csv

Rows × Columns: 10 × 21

Primary Key: Id

Foreign Keys: None

Column	Data Type	Description
Id	<i>string (UUID)</i>	Primary key. Unique payer identifier.
NAME	<i>string</i>	Payer/insurance company name.
ADDRESS	<i>string</i>	Headquarters address.
CITY	<i>string</i>	Headquarters city.
STATE_HEADQUARTERED	<i>string</i>	Headquarters state.
ZIP	<i>float</i>	Headquarters ZIP code.
PHONE	<i>string</i>	Contact phone number.
AMOUNT_COVERED	<i>float</i>	Total dollar amount this payer has covered.
AMOUNT_UNCOVERED	<i>float</i>	Total dollar amount left uncovered (patient responsibility).
REVENUE	<i>float</i>	Total revenue collected by the payer.
COVERED_ENCOUNTERS	<i>int</i>	Count of encounters covered.
UNCOVERED_ENCOUNTERS	<i>int</i>	Count of encounters not covered.
COVERED_MEDICATIONS	<i>int</i>	Count of medications covered.
UNCOVERED_MEDICATIONS	<i>int</i>	Count of medications not covered.
COVERED_PROCEDURES	<i>int</i>	Count of procedures covered.
UNCOVERED_PROCEDURES	<i>int</i>	Count of procedures not covered.
COVERED_IMMUNIZATIONS	<i>int</i>	Count of immunizations covered.
UNCOVERED_IMMUNIZATIONS	<i>int</i>	Count of immunizations not covered.
UNIQUE_CUSTOMERS	<i>int</i>	Distinct patients ever covered by this payer.
QOLS_AVG	<i>float</i>	Average Quality-of-Life Score across covered patients.
MEMBER_MONTHS	<i>int</i>	Total member-months of enrollment.

5. ENCOUNTERS

One row per patient visit/encounter. The central table that CONDITIONS, PROCEDURES, MEDICATIONS, and OBSERVATIONS all tie back to.

Source file: encounters.csv

Rows × Columns: 61,459 × 15

Primary Key: Id

Foreign Keys: PATIENT → PATIENTS.Id | PROVIDER → PROVIDERS.Id | PAYER → PAYERS.Id

Column	Data Type	Description
Id	<i>string (UUID)</i>	Primary key. Unique encounter identifier.
START	<i>datetime</i>	Encounter start timestamp.
STOP	<i>datetime</i>	Encounter end timestamp.
PATIENT	<i>string (UUID)</i>	Foreign key to PATIENTS.Id.
ORGANIZATION	<i>string (UUID)</i>	Facility where encounter occurred (see ORGANIZATIONS.Id).
PROVIDER	<i>string (UUID)</i>	Foreign key to PROVIDERS.Id — clinician seen.
PAYER	<i>string (UUID)</i>	Foreign key to PAYERS.Id — insurer at time of encounter.
ENCOUNTERCLASS	<i>string</i>	Encounter type (ambulatory, emergency, inpatient, wellness, urgentcare).
CODE	<i>int</i>	SNOMED-CT code for the encounter type.
DESCRIPTION	<i>string</i>	Human-readable description of the encounter.
BASE_ENCOUNTER_COST	<i>float</i>	Base cost of the encounter (excludes procedures/meds).
TOTAL_CLAIM_COST	<i>float</i>	Total cost claimed for the encounter.
PAYER_COVERAGE	<i>float</i>	Amount covered by the payer.
REASONCODE	<i>float</i>	SNOMED-CT code for the reason for the encounter.
REASONDESCRIPTION	<i>string</i>	Description of the reason for the encounter.

6. CONDITIONS

Diagnoses / medical conditions recorded for a patient during an encounter. Feeds into CAREPLANS via REASONCODE.

Source file: conditions.csv

Rows × Columns: 38,094 × 6

Primary Key: None (event table)

Foreign Keys: PATIENT → PATIENTS.Id | ENCOUNTER → ENCOUNTERS.Id

Column	Data Type	Description
START	date	Date the condition was diagnosed/onset.
STOP	date	Date the condition resolved; blank if ongoing.
PATIENT	string (UUID)	Foreign key to PATIENTS.Id.
ENCOUNTER	string (UUID)	Foreign key to ENCOUNTERS.Id.
CODE	int	SNOMED-CT diagnosis code.
DESCRIPTION	string	Human-readable condition description.

7. PROCEDURES

Clinical procedures performed on a patient during an encounter.

Source file: procedures.csv

Rows × Columns: 83,823 × 9

Primary Key: None (event table)

Foreign Keys: PATIENT → PATIENTS.Id | ENCOUNTER → ENCOUNTERS.Id

Column	Data Type	Description
START	<i>datetime</i>	Procedure start timestamp.
STOP	<i>datetime</i>	Procedure end timestamp.
PATIENT	<i>string (UUID)</i>	Foreign key to PATIENTS.Id.
ENCOUNTER	<i>string (UUID)</i>	Foreign key to ENCOUNTERS.Id.
CODE	<i>int</i>	SNOMED-CT procedure code.
DESCRIPTION	<i>string</i>	Human-readable procedure description.
BASE_COST	<i>float</i>	Cost of the procedure.
REASONCODE	<i>float</i>	SNOMED-CT code for reason performed.
REASONDESCRIPTION	<i>string</i>	Description of reason performed.

8. MEDICATIONS

Medications prescribed/dispensed to a patient, with cost and payer coverage detail.

Source file: medications.csv

Rows × Columns: 56,430 × 13

Primary Key: None (event table)

Foreign Keys: PATIENT → PATIENTS.Id | ENCOUNTER → ENCOUNTERS.Id | PAYER → PAYERS.Id

Column	Data Type	Description
START	date	Date medication was started.
STOP	date	Date medication was stopped; blank if ongoing.
PATIENT	string (UUID)	Foreign key to PATIENTS.Id.
PAYER	string (UUID)	Foreign key to PAYERS.Id — insurer billed.
ENCOUNTER	string (UUID)	Foreign key to ENCOUNTERS.Id.
CODE	int	RxNorm medication code.
DESCRIPTION	string	Medication name/description.
BASE_COST	float	Base cost per dispense.
PAYER_COVERAGE	float	Amount covered by the payer.
DISPENSES	int	Number of times the medication was dispensed.
TOTALCOST	float	Total cost across all dispenses.
REASONCODE	float	SNOMED-CT code for reason prescribed.
REASONDESCRIPTION	string	Description of reason prescribed.

9. OBSERVATIONS

Clinical measurements and lab results (vitals, labs, survey responses) recorded during an encounter. Largest table in scope.

Source file: observations.csv

Rows × Columns: 531,144 × 9

Primary Key: None (event table)

Foreign Keys: PATIENT → PATIENTS.Id | ENCOUNTER → ENCOUNTERS.Id

Column	Data Type	Description
DATE	<i>datetime</i>	Date/time observation was recorded.
PATIENT	<i>string (UUID)</i>	Foreign key to PATIENTS.Id.
ENCOUNTER	<i>string (UUID)</i>	Foreign key to ENCOUNTERS.Id.
CATEGORY	<i>string</i>	Observation category (vital-signs, laboratory, survey, etc.).
CODE	<i>string</i>	LOINC code identifying the observation type.
DESCRIPTION	<i>string</i>	Human-readable observation name.
VALUE	<i>string</i>	Recorded value (numeric or text).
UNITS	<i>string</i>	Unit of measure for VALUE.
TYPE	<i>string</i>	Value datatype (numeric, text, coded).

10. CAREPLANS

Structured care plans (e.g., diabetes self-management) assigned to a patient, typically linked to a condition via REASONCODE.

Source file: careplans.csv

Rows × Columns: 3,931 × 9

Primary Key: Id

Foreign Keys: PATIENT → PATIENTS.Id | ENCOUNTER → ENCOUNTERS.Id

Column	Data Type	Description
Id	string (UUID)	Primary key. Unique care plan identifier.
START	date	Care plan start date.
STOP	date	Care plan end date; blank if ongoing.
PATIENT	string (UUID)	Foreign key to PATIENTS.Id.
ENCOUNTER	string (UUID)	Foreign key to ENCOUNTERS.Id.
CODE	int	SNOMED-CT care plan code.
DESCRIPTION	string	Care plan description.
REASONCODE	float	SNOMED-CT code for the condition prompting the plan (relates to CONDITIONS.CODE).
REASONDESCRIPTION	string	Description of the reason for the care plan.

Appendix A: Relationship Matrix

Every parent-child relationship in the Core ER Diagram, in one-to-many (crow's foot) form. The last row is a logical relationship inferred from matching code values, not a declared foreign key.

Parent	Child	Relationship	Notes
Organizations	Providers	<i>One-to-Many</i>	Organization employs many providers
Organizations	Encounters	<i>One-to-Many</i>	Encounter occurs at one organization
Providers	Encounters	<i>One-to-Many</i>	Provider performs many encounters
Patients	Encounters	<i>One-to-Many</i>	Patient has many encounters
Payers	Encounters	<i>One-to-Many</i>	Payer covers many encounters
Encounters	Conditions	<i>One-to-Many</i>	Encounter diagnoses many conditions
Encounters	Procedures	<i>One-to-Many</i>	Encounter includes many procedures
Encounters	Medications	<i>One-to-Many</i>	Encounter prescribes many medications
Encounters	Observations	<i>One-to-Many</i>	Encounter records many observations
Encounters	Careplans	<i>One-to-Many</i>	Encounter initiates many careplans
Conditions	Careplans	<i>Logical (not a true FK)</i>	Careplans.REASONCODE relates to Conditions.CODE